An Epidemiological Modeling Take on Religion Dynamics

Bilge Taskin*[1] and Teddy Lazebnik*†[1,2]

[1]Department of Computing, Jönköping University, Jönköping, Sweden
[2]Department of Information Systems, University of Haifa, Haifa, Israel

Abstract

We propose an epidemiologically inspired model of religious change that distinguishes passive believers, active missionaries, and a protected institutional role within competing affiliations. Conversion, disaffiliation, demographic turnover, and missionary relabeling are represented by a fixed-dimensional compartmental system and a corresponding stochastic process. Controlled experiments distinguish schism-mediated replacement, externally forced transient movements, and locally stable mutation-supported coexistence. Ablations identify the mechanisms required for each example, while a continuous sensitivity map exposes substantial dependence on the observation horizon. Separately, reduced descriptive models are fitted to Swedish church membership, Turkish Jehovah's Witness publisher activity, and New Zealand no-religion reporting. A continuous rate shift improves the Swedish fit, but linear trends outperform the selected diffusion curves on the Turkish and New Zealand temporal holdouts. These historical fits do not identify the full multi-religion mechanism or causal social and political effects. The contribution is a reproducible framework for formulating and testing hypotheses about religious diffusion, together with explicit boundaries between mechanistic simulation, imposed forcing, and aggregate curve fitting.

Keywords: religious dynamics; multi-strain diffusion; mathematical sociology; compartmental modeling; schism.

1 Introduction

Religion has played a central role in shaping human societies, influencing cultural identity, political institutions, and collective behavior throughout history [1, 2, 3]. In early agricultural societies, animistic and polytheistic traditions provided explanatory frameworks for natural phenomena and mechanisms for social cohesion [4, 5, 6]. The institutionalization of priesthoods and ritual practices in Mesopotamia, Egypt, and the Indus Valley can be seen as early examples of religion serving as a stabilizing force for centralized states, legitimizing political authority and codifying moral order [7, 8, 9]. Later, the emergence of axial-age religions such as Judaism, Buddhism, Confucianism, and early Greek philosophy introduced universal moral codes and abstract cosmologies, enabling broader social integration and the consolidation of empires [10, 11, 12]. Following this in the historical record, in the classical and medieval periods, the spread of Christianity and Islam illustrates how religious movements can transcend ethnic and geographic boundaries, often

*These authors contributed equally.
†Corresponding author: teddy.lazebnik@ju.se.

accompanying processes of imperial expansion, trade, and conquest [13, 14, 15]. Religious institutions became central to governance, education, and law, shaping not only the political order but also economic practices such as taxation, charity, and land distribution [16, 17]. Nowadays, modernity has not diminished religion's influence, but transformed its modalities of spread and competition. The Enlightenment and secularization movements challenged traditional institutions, yet religion adapted by giving rise to new denominations, evangelical movements, and, in some cases, politically mobilized fundamentalisms [18, 19, 20]. In the globalized era, religions circulate alongside cultural products and ideas, facilitated by digital media and migration [21, 22]. Contemporary dynamics include the expansion and diversification of Pentecostal and charismatic movements across national boundaries [23, 24, 25]. At the same time, disaffiliation and nonreligious identification have emerged as significant counter-trends in many industrialized societies [26].

These historical patterns strongly suggest that religions function analogously to evolving systems of ideas that compete for adherents and adapt to shifting socio-political environments [27]. Crucially, religion not only spreads within societies but also reshapes them, influencing demographic patterns, institutional stability, and economic organization [28]. For this reason, the spread of religion has long been a subject of scholarly attention across disciplines [29, 30, 31, 32]. Historians and sociologists have examined large-scale processes of conversion, conflict, and reform; anthropologists have investigated the ritual, symbolic, and cultural mechanisms that facilitate belief transmission; and political scientists have studied the role of religion in shaping governance, legitimacy, and social movements.

Alongside this scholarship, formal models have examined church growth, new religious movements, evolutionary diversification, and changes in religiosity [33, 34, 35, 36]. These studies motivate examining how competition, behavioral roles, and schism interact within one transparent dynamical structure. To address this limitation, recent research has drawn attention to the structural similarities between the spread of religions and the spread of infectious diseases [33, 37]. Both involve populations of susceptible individuals, mechanisms of contact that facilitate transmission, processes of "infection" (conversion), and "recovery" (disaffiliation or apostasy). Furthermore, just as pathogens may coexist, compete, or mutate, so too can religions diversify, fragment, and evolve.

This analogy naturally motivates borrowing compartmental mechanisms from epidemiology, including the Susceptible–Infected–Recovered framework [38], without assuming that religious affiliation follows literal disease states. In this work, we introduce a multi-religion model distinguishing passive believers, active missionaries, and a protected institutional role. Conversion, disaffiliation, competition, and schism-like relabeling can generate different trajectories under explicitly specified settings. We also define an optional link between external covariates and conversion rates, while keeping that untested extension separate from the experiments actually performed.

This paper makes three primary contributions. First, we present a role-structured diffusion model with a fixed set of competing affiliations. Second, we provide reproducible deterministic and stochastic experiments, mechanism ablations, and a local coexistence analysis that distinguish endogenous schism from externally forced transients. Third, we assess what reduced historical curve fits can and cannot establish, using documented observation definitions and temporal holdouts. These fits illustrate aggregate descriptions; they do not calibrate or validate the full multi-religion framework or identify causal social and political effects.

The remainder of this paper is organized as follows. Section 2 reviews cultural and religious

diffusion and relevant epidemiological models. Section 3 defines the states, transitions, and optional contextual-rate function. Section 4.1 describes the simulations, historical observations, and sensitivity design. Section 4.2 reports the resulting trajectories, empirical fitting and holdout performance, and numerical agreement. Section 5 discusses interpretation, limitations, and future research; the Appendix provides computational and calibration details.

2 Related Work

In this section, we review the underlying religion spread dynamics that one should consider in any modeling effort, followed by a short review of epidemiological modeling practices with a focus on extended SIR models with multi-strain.

2.1 Religion spread dynamics

Religion has long been recognized as one of the most enduring and influential forces in human societies [39, 40, 41]. It provides not only systems of belief but also shared practices, moral codes, and institutional structures that bind communities together and regulate social behavior [42]. Scholars across sociology, anthropology, history, and political science have examined religion as both a driver and a product of social change, noting its capacity to shape collective identities, stabilize political orders, and motivate large-scale cooperation [43, 44, 45, 46]. At the same time, religion has been a source of competition, division, and conflict, often catalyzing transformative cultural and political shifts [47, 48].

The spread of religion is not simply a matter of individual belief but a complex socio-cultural process influenced by institutions, social networks, and political environments. Early anthropological theories emphasized religion as a form of "social glue", binding communities through ritual and shared narratives [49]. In agricultural and early state societies, religious practices often reinforced stratified political hierarchies: priesthoods legitimized rulers, while rulers endowed temples, intertwining spiritual authority with political power [50]. The integration of religion into law and governance provided durable mechanisms for transmitting norms across generations and stabilizing expanding polities [51].

At a macro-historical scale, religion often spread through contact processes that resemble contagion: through trade, conquest, migration, and missionary activity [21]. The expansion of Buddhism along the Silk Road, Christianity throughout the Roman Empire, and Islam across North Africa and into Asia are notable examples where the growth of religious communities followed routes of human mobility and exchange [52, 53]. In such contexts, social prestige, institutional support, and political patronage frequently amplified the rate of conversion, while persecution and legal restrictions could slow or suppress diffusion. Thus, the likelihood of religious adoption was never purely a matter of theology but deeply conditioned by social and political context [29].

Another key feature of religious spread is competition among belief systems. Rarely do religions exist in isolation; instead, they compete for adherents within overlapping populations. This competition can manifest as coexistence (as in multi-religious empires), dominance by a single tradition (as in state religions), or dynamic cycles of conversion and reform. Religious competition has historically led to both violent conflict (such as the Crusades or sectarian wars) and peaceful coexistence under frameworks of tolerance or pluralism [54, 55, 56, 57]. Notably, this competitive aspect mirrors ecological and epidemiological systems, where multiple strains or

species vie for limited resources.

Religions also split and diversify, creating new sects, denominations, or entirely new movements [58, 59]. Schisms within Christianity, the formation of Sunni and Shia traditions in Islam, or the branching of Hindu and Buddhist schools illustrate how internal doctrinal disputes, leadership crises, or external pressures can fragment a religion into competing sub-groups. These splits often lead to rapid bursts of growth for the new branch, particularly when aligned with favorable social or political conditions. Such processes resemble mutation in biological systems, generating diversity within a lineage that can alter the competitive balance among strains.

Growth, institutionalization, fragmentation, and decline are additional patterns in religious history [58, 59]. New religious movements often begin as small, tightly bound communities, sometimes dismissed as cults or sects, but may grow rapidly when their message resonates with marginalized or disaffected groups [60]. Over time, some of these movements institutionalize and become dominant traditions, while others decline, fragment, or disappear altogether. The rise-and-fall cycles of millenarian cults, revivalist movements, and reformist sects highlight the temporal dynamics of religious spread and decline, often linked to broader socio-political turbulence.

Taken together, these insights point toward a few essential features of religion spread that are particularly relevant for formal modeling. First, religions diffuse through contact-like processes, influenced by networks of social interaction and mobility. Second, they compete as multiple strains within the same population, where outcomes can include coexistence, dominance, or extinction. Third, religions are subject to mutation-like splitting, producing new branches that alter the competitive landscape. Fourth, their dynamics can exhibit rise-and-fall patterns, reflecting interactions between belief systems and broader socio-political forces. Finally, the rate and direction of spread are strongly modulated by external conditions—such as political stability, institutional regulation, and cultural diversity—which motivate explicit contextual extensions and appropriate empirical tests.

2.2 Epidemiological modeling

There are many types of modeling methods suggested in the literature to capture pandemic spread dynamics [61, 62, 63]. The simplest type of models are functional (phenomenological) models, which include logistic, Richards, Gompertz, and other growth curves fitting [64]. Other models focus on a bottom-to-top approach, simulating individual-level interactions to capture population-level spread dynamics with methods like the agent-based simulation [65, 66, 67, 68]. Data-driven methods also include physics-informed neural networks that combine differential-equation structure with observed trajectories [69]. Nevertheless, compartment models, such as the SIR model, remain widely used because their compartmental mechanisms are explicit and can be confronted with observations [70, 71]. In these models, one partitions the population into epidemiological states and describes flows between them via differential (or difference) equations [38].

A foundational example is the classical SIR model, which decomposes the population into three compartments: $S(t)$, $I(t)$, and $R(t)$ with dynamics described by the following system of ordinary differential equations (ODEs):

$$\frac{dS}{dt} = -\beta \frac{SI}{N}, \quad \frac{dI}{dt} = \beta \frac{SI}{N} - \gamma I, \quad \frac{dR}{dt} = \gamma I, \tag{1}$$

where $N = S + I + R$ is constant, $\beta > 0$ is the effective transmission rate, and $\gamma > 0$ is the recovery/removal rate [38]. This model is deliberately simplified and does not capture all real-world epidemiological processes. As such, many extensions have been proposed over the years that extend it to capture more realistic epidemiological dynamics. Most extensions are obtained by relaxing its simplifying assumptions, so it is useful to view them systematically. Common extensions: (i) add infection stages (e.g., exposed/latent, presymptomatic, asymptomatic) to represent progression and infection-to-infectiousness delays [72]; (ii) distinguish outcomes of infection (e.g., hospitalization, severe disease, death) as separate compartments rather than absorbing them into a single removed class [72]; (iii) introduce additional time scales via demographic turnover, waning immunity, or time-varying transmission such as seasonality [72]; and (iv) relax homogeneous mixing by incorporating population structure (age, risk, space), yielding coupled compartmental systems [73].

These mechanisms can first be developed for a single pathogen and then extended to competing strains. Real-world pandemics frequently involve multiple co-circulating strains (or variants) whose epidemiological parameters and immune-escape properties differ over time [74]. Extending single-strain compartmental models to multi-strain settings can be done at several levels of resolution, ranging from minimal "parallel-strain" models to history-based constructions that track immune states induced by prior infections.

A generic strategy is to represent strain identity explicitly (e.g., by duplicating infected compartments) and to encode immunological interactions using either (i) complete cross-immunity, (ii) partial cross-immunity through a susceptibility reduction matrix, or (iii) explicit immune-history compartments indexed by the set (or sequence) of previously experienced strains [75].

The most parsimonious multi-strain extension assumes that infection by any strain removes an individual from future susceptibility (i.e., no reinfection by any strain). Let $I_i(t)$ denote the number infectious with strain $i \in \{1, \dots, m\}$, with strain-specific transmission and recovery rates (β_i, γ_i). A natural model is

$$\begin{aligned} \frac{dS}{dt} &= -\frac{S}{N}\sum_{i=1}^{m} \beta_i I_i, \\ \frac{dI_i}{dt} &= \beta_i \frac{S I_i}{N} - \gamma_i I_i, \qquad i = 1, \dots, m, \\ \frac{dR}{dt} &= \sum_{i=1}^{m} \gamma_i I_i, \end{aligned} \tag{2}$$

where $N = S + \sum_i I_i + R$ is constant. This formulation captures competition for a shared susceptible pool and represents competition during a transient outbreak, but does not by itself establish persistent endemic coexistence. Models that relax complete cross-protection can instead track partial cross-immunity and infection history [76, 77, 78].

Allowing reinfection requires relaxing the assumption that recovery permanently removes individuals from the susceptible class. Two common mechanisms are waning immunity and incomplete cross-immunity. In the first, individuals who recovered return to susceptibility over time while in the latter, prior infection reduces but does not eliminate susceptibility to the same or other strains. A convenient reduced description introduces strain-specific recovered classes R_i and permits infection of previously recovered individuals at a reduced rate. For example, letting $\sigma_{ij} \in [0, 1]$ denote the relative susceptibility to strain j after recovery from strain i, a schematic reinfection term takes the form $\beta_j \sigma_{ij} R_i I_j / N$. In its most general form, reinfection

leads to immune-history indexing (e.g., R_J for individuals recovered from the set of strains J), which scales combinatorially with the number of strains and naturally encodes infection order [75]. Such models are useful when repeated infections materially affect population-level dynamics, including thresholds in which reinfection becomes dominant [79], strain invasion and coexistence regimes [80], and settings where a newly emerging strain can effectively infect individuals with prior immunity [81].

Multi-strain models can treat strain appearance as exogenous where new strains are introduced at specified times or endogenous where new strains are generated by mutation from existing strains. A common endogenous representation is a mutation flow between infectious classes:

$$\frac{dI_i}{dt} = \cdots - \sum_{j \neq i} \mu_{ij} I_i + \sum_{j \neq i} \mu_{ji} I_j, \tag{3}$$

where μ_{ij} is the rate at which infections of strain i generate infections of strain j (e.g., via within-host mutation and onward transmission). For many-strain settings, mutation is often coupled to antigenic distance so that both (a) mutation transitions and (b) cross-immunity depend on a strain-space metric. This yields a bridge between discrete multi-strain compartment models and evolutionary/antigenic-drift descriptions [82, 83, 84].

3 Model Definition

We model religious affiliation as a multi-strain, contact-driven diffusion process in which religions compete for adherents. The epidemiological analogy specifies transition mechanisms; it does not identify religious belief with pathology. The model combines role heterogeneity, cross-affiliation conversion, demographic turnover, and missionary relabeling representing schism within a fixed set of potential religions [75, 74]. Unlike classical SIR, disaffiliation returns individuals to a susceptible state, and there is no permanently recovered compartment. Figure 1 summarizes the transitions.

Let $\mathcal{R} = \{1, \ldots, m\}$ be a fixed set of modeled religions; some may initially be absent. Each person occupies exactly one state: S (unaffiliated), B_r (affiliated but not actively proselytizing), M_r (actively proselytizing), or P_r (an institutional or religious elite protected from direct cross-conversion). Thus

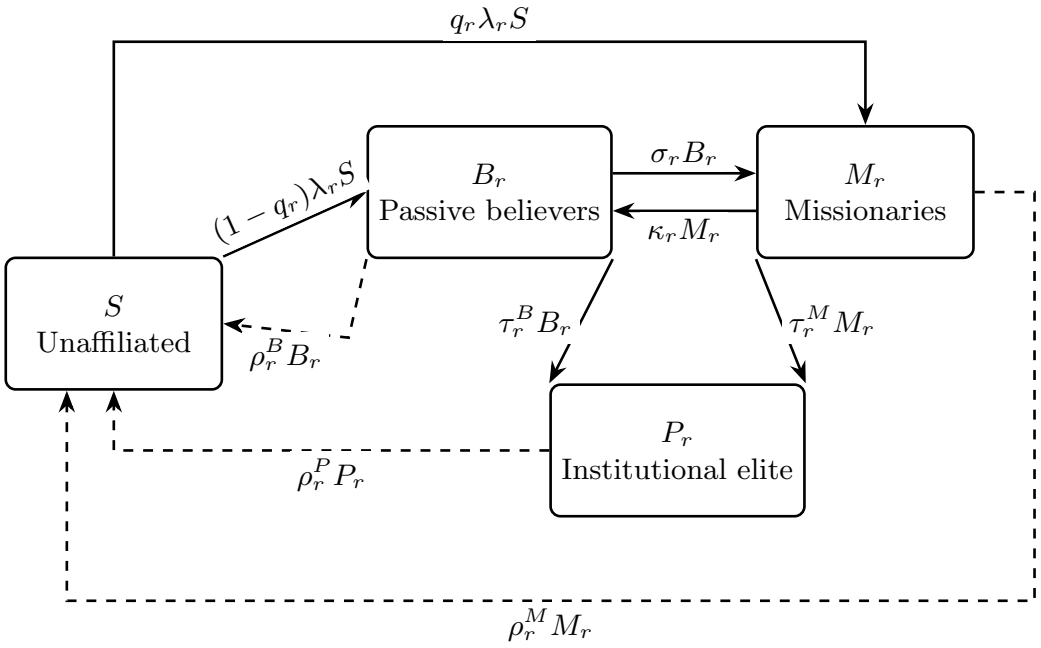
$$N(t) = S(t) + \sum_{r=1}^{m} [B_r(t) + M_r(t) + P_r(t)]. \tag{4}$$

The model excludes simultaneous affiliations. The distinction between B_r, M_r, and P_r is behavioral and institutional, not a measured distinction in the historical datasets used below.

Under homogeneous mixing, the force of conversion to religion r is

$$\lambda_r(t) = \beta_r(t) \frac{M_r(t)}{N(t)}, \qquad X_r(t) = S(t) + \sum_{j \neq r} [B_j(t) + M_j(t)]. \tag{5}$$

Here β_r is an effective contact-and-conversion rate per missionary, not a probability conditional on a measured contact. All continuous-time rates have units year^{-1}; $q_r \in [0, 1]$ is dimensionless. Conversion into religion r occurs at rate $\lambda_r X_r$ and is divided between B_r and M_r with weights $1 - q_r$ and q_r. Each B_r and M_r individual faces an outward cross-conversion hazard $\sum_{\ell \neq r} \lambda_\ell$. Priests do not directly generate conversions and cannot be directly converted to another religion,



Coupling to other religions

Cross-conversion: $B_r, M_r \to B_\ell$ or M_ℓ at hazard λ_ℓ; $\ell \neq r$.
Destination roles are selected with probabilities $1 - q_\ell$ and q_ℓ.

Schism: $M_r \to M_\ell$ at hazard $\nu_{r\ell}$. No direct cross-conversion of P_r.

Demography: births bX and deaths μX in every compartment X.

Figure 1: Role-structured transitions for a focal religion r. Conversion is generated by missionaries; a new convert enters B_r with probability $1 - q_r$ or M_r with probability q_r. Cross-conversion and missionary relabeling couple this structure to the other religions, as specified below the diagram. Births and deaths apply to every compartment.

although they may disaffiliate.

Within an affiliation, σ_r governs $B_r \to M_r$, κ_r governs $M_r \to B_r$, and τ_r^B, τ_r^M govern progression into P_r. Disaffiliation from role $X \in \{B, M, P\}$ occurs at rate ρ_r^X. Missionary relabeling from r to ℓ occurs at rate $\nu_{r\ell} \geq 0$, with $\nu_{rr} = 0$. This transfers an existing individual from M_r to M_ℓ; it does not create a person or an additional equation. It represents entry into a previously specified denomination, rather than unrestricted innovation of new labels.

Births and deaths occur at rates bX and μX in every compartment X. This composition-preserving assumption is a demographic bookkeeping approximation: it does not assert that newborns literally become missionaries or priests. It implies $\dot{N} = (b-\mu)N$. Uniform demographic growth cancels from the deterministic equations for population fractions, although demographic sampling contributes finite-population variation in the stochastic implementation.

The resulting ODE system is

$$\dot{S} = (b-\mu)S - S\sum_k \lambda_k + \sum_k (\rho_k^B B_k + \rho_k^M M_k + \rho_k^P P_k), \tag{6}$$

$$\dot{B}_r = (b-\mu)B_r + (1-q_r)\lambda_r X_r - B_r \sum_{\ell \neq r} \lambda_\ell - \sigma_r B_r + \kappa_r M_r - (\rho_r^B + \tau_r^B)B_r, \tag{7}$$

$$\begin{aligned}\dot{M}_r = {} & (b-\mu)M_r + q_r\lambda_r X_r - M_r \sum_{\ell \neq r} \lambda_\ell + \sigma_r B_r - (\kappa_r + \rho_r^M + \tau_r^M)M_r \\ & + \sum_{\ell \neq r} \nu_{\ell r} M_\ell - \sum_{\ell \neq r} \nu_{r\ell} M_r,\end{aligned} \tag{8}$$

$$\dot{P}_r = (b-\mu)P_r + \tau_r^B B_r + \tau_r^M M_r - \rho_r^P P_r. \tag{9}$$

All transfers are nonnegative at the boundary of the state space, so nonnegative initial states remain nonnegative. Summing the equations cancels conversion, role changes, disaffiliation, and mutation. In particular, an absent religion with no incoming missionary relabeling and no external introduction remains absent. This property supplies a useful implementation check and distinguishes endogenous seeding from imposed introductions.

Optional contextual modulation. To make the proposed connection to external conditions explicit, a positive-rate extension is

$$\beta_r(t) = \beta_{r,0} \exp\{\boldsymbol{\eta}_r^\mathsf{T} \mathbf{z}_r(t)\}, \tag{10}$$

where $\mathbf{z}_r(t)$ contains specified social, institutional, or political covariates and $\boldsymbol{\eta}_r$ their coefficients. The implementation accepts supplied covariate series, with linear interpolation between supplied times. This extension is motivated by context-sensitive social-epidemiological modeling [85]; it is not estimated or empirically tested here. The experiments instead use constant rates, or explicitly prescribed piecewise-constant rates in Scenario B and the reduced Sweden fit. No measured political effect, or causal interpretation of a fitted rate change, is claimed.

Role of the historical reduction. For a single religion, setting $B = P = 0$, $q = 1$, all role-transition rates and mutation to zero, and $y = M/N$ gives exactly

$$\dot{y} = \beta(t)y(1-y) - \rho(t)y. \tag{11}$$

These restrictive assumptions remove the mechanisms that distinguish the full model. More generally, $\dot{y}$ depends on the unobserved missionary share and role-specific losses, so aggregate affiliation alone does not close the full system. We therefore use Eq. (11) as a descriptive reduced benchmark, not as empirical validation of role heterogeneity, competition, or schism. For publisher activity and no-religion reporting, its coefficients are phenomenological rather than literal missionary conversion and apostasy rates.

4 Model Analysis

4.1 Experimental design

We distinguish three tasks: controlled simulations of the full model, descriptive historical fits of reduced curves, and a finite-horizon sensitivity map. All simulated rates and horizons are expressed in years. The stylized parameterizations are illustrative, not calibrated historical time scales. We use $N_0 = 10{,}000$, $b = \mu = 0.01$, and stochastic step $\Delta t = 0.1$ year. Each main scenario has 100 independent replicates. The deterministic system is solved with adaptive DOP853, relative and absolute tolerances 10^{-9}, with integration restarted from the evolved endpoint at every scheduled event. Details of the exchangeable-agent stochastic implementation, seeds, software versions, and numerical checks appear in the Appendix.

4.1.1 Stylized religious dynamics

Table 1: Executable behavioral parameters. All entries except q have units year^{-1}. B's last two columns give each minor group's rates before and after its own scheduled change. Unlisted demographic settings are $b = \mu = 0.01$ in all scenarios.

Parameter	A: $r = 1$	A: $r = 2$	B: $r = 1$	B: minor, pre	B: minor, post	C: both
β	0.03	0.45	0.04	0.75	0.02	0.28
q	0.05	0.1	0.05	0.02	0.02	0.05
σ	0.02	0.03	0.02	0.005	0.005	0.018
κ	0.01	0.01	0.02	0.02	0.02	0.018
τ^B	0.002	0.003	0.002	0.0003	0.0003	0.0015
τ^M	0.003	0.004	0.003	0.0003	0.0003	0.002
ρ^B	0.005	0.004	0.006	0.01	0.05	0.006
ρ^M	0.004	0.003	0.006	0.01	0.15	0.006
ρ^P	0.001	0.001	0.001	0.01	0.4	0.001

Table 2: Initial counts and coupling. Counts for B are specified immediately before its year-zero introduction. All scenarios start with $N_0 = 10{,}000$, and all omitted mutation entries are zero.

Scenario	S_0	(B_1, M_1, P_1)	(B_2, M_2, P_2)	T (years)	Mutation (year^{-1})
A	6000	$(3000, 200, 800)$	$(0, 0, 0)$	216	$\nu_{12} = 0.002$
B	5000	$(4200, 300, 500)$	$(0, 0, 0)$	216	None
C	6300	$(2000, 120, 180)$	$(1200, 80, 120)$	1000	$\nu_{12} = \nu_{21} = 0.02$

Scenario A (schism-mediated emergence and replacement). Religion 1 is the only initially present religion, with 40% of the population affiliated; 60% is unaffiliated. Religion 2

starts absent and is seeded solely by missionary relabeling at $\nu_{12} = 0.002$ year^{-1}. Its conversion rate is 0.45, compared with 0.03 for religion 1. All other settings are listed in Tables 1–2. We simulate 216 years and compare the result with a no-schism ablation that changes only ν_{12} to zero.

Scenario B (externally introduced, forced transient movements). Five initially absent minor religions are introduced sequentially by transfers from S, without mutation. Each begins with 50 passive believers and 100 missionaries. At its separately prescribed change time, its conversion rate decreases from 0.75 to 0.02, while its disaffiliation rates increase (Table 1). Table 3 specifies every event. This experiment asks whether the same compartmental structure represents growth followed by decline under explicit external forcing; it does not test spontaneous emergence or endogenous recurrent waves. Ablations remove either all introductions or all subsequent rate changes.

Table 3: Scenario B's externally prescribed events. Each introduction transfers 50 people from S to B_r and 100 from S to M_r, preserving N. At the change year, only the four rates indicated in Table 1 change.

Minor religion	Introduction year	Rate-change year
2	0	40
3	16	60
4	48	90
5	112	145
6	167	190

Scenario C (symmetric, mutation-supported coexistence). The religions have identical behavioral parameters but unequal initial prevalence. Symmetric missionary relabeling is $\nu_{12} = \nu_{21} = 0.02$ year^{-1}. We integrate to 1000 years to examine relaxation, not to make a millennium-scale social forecast. We locate a coexistence equilibrium and evaluate the Jacobian after eliminating S using the population-sum constraint. Negative real parts of all six remaining eigenvalues establish local, not global, asymptotic stability. A separate ablation reduces both relabeling rates to 0.0002; symmetry of behavioral parameters is not assumed sufficient for balanced coexistence.

4.1.2 Historical records and observation definitions

Sweden (membership decline). We use the Church of Sweden's official membership percentages for 1972–2024, excluding 1981, for which the source has no population-register figure: 52 observations in total [86]. The observed variable is membership divided by population, not the missionary compartment. A constant-rate reduced model is compared with a four-rate specification using a fixed breakpoint at 2000. The breakpoint is a prespecified descriptive comparison; its timing does not identify the effect of any particular institutional change. The source also records changes in membership rules, underscoring that these rates aggregate more than interpersonal conversion.

Turkey (active publisher reporting). We use annual peak Jehovah's Witness publisher counts divided by the population reported in the corresponding service-year table, 2014–2024 (11 observations) [87, 88, 89, 90]. Peak and annual-average publisher counts are not mixed. These organizational reports measure active publishing rather than all adherents, so they are not observations of $B + M + P$. We fit the two-parameter logistic description $\dot{y} = ry(1 - y/K)$,

where r is an effective growth rate and K is an asymptotic share. We do not separately fit three redundant parameters in $\beta y(K-y)-\rho y$, since that equation identifies at most $\beta K-\rho$ and β from an ideal trajectory. We compare the logistic curve with anchored linear and exponential trends, without an additional fitted breakpoint.

New Zealand (no-religion reporting). We assemble no-religion counts for the 2001, 2006, 2013, 2018, and 2023 censuses from Stats NZ releases [91, 92]. We divide each by the corresponding *all-resident census count*, not a sum of potentially overlapping affiliation categories or a mixture of response-based denominators [93, 94, 95, 92]. Table 7 supplies all five numerators and denominators. Consequently, the complement includes unclassified or nonresponding residents as well as people reporting a religion. This is a descriptive two-category reporting series, not a closed observation of religious versus nonreligious switching. Common denominator definitions do not eliminate changes in response, imputation, or census processing. We therefore also fit the separately retained published percentage series as a sensitivity check, without mixing it into the primary count ratio. No breakpoint is fitted to five observations.

Fitting and temporal holdouts. All curves start at the first observed value. Dynamical parameters are fitted by bounded nonlinear least squares, using 32 deterministic multi-start initializations for the nonlinear two- and four-parameter fits. Exact continuous segment solutions replace numerical integration for these reduced equations. The Sweden state is propagated continuously through 2000 and is never reset to an observed breakpoint value. Because the first point is fixed, all reported fitting metrics exclude it, leaving 51, 10, and 4 informative residuals respectively. An anchored linear trend has one fitted slope; the Turkish exponential comparison has one fitted rate. All comparisons minimize squared errors on the original share scale. Bounds, diagnostic AICc values, and the exact solution are given in the Appendix.

We also fit only observations through 2014 for Sweden, 2021 for Turkey, and 2018 for New Zealand, and evaluate the remaining 10, 3, and 1 observations without updating parameters or reinitializing states. These are illustrative historical holdouts, not prospective preregistered forecasts. Their small number, especially one NZ point, limits generalization. No uncertainty intervals for historical mechanism parameters are asserted; boundary solutions and sensitivity to the reporting definition are reported rather than interpreted as precise causal estimates.

4.1.3 Finite-horizon sensitivity

We vary Scenario A's β_2 over 21 equally spaced values in $[0.03, 0.60]$ and ρ_2^M over 21 values in $[0.001, 0.15]$, keeping all other settings fixed. Each of the 441 configurations is solved deterministically. We report the continuous share $y_2(216)$ and retain $y_2(432)$ to assess horizon dependence. There are no threshold-assigned "takeover" or "coexistence" labels, and the map does not estimate takeover probabilities or establish a bifurcation.

4.2 Results

4.2.1 Full-model simulations and mechanism checks

Figure 2 shows mean trajectories with pointwise 2.5th–97.5th percentiles across stochastic runs. In Scenario A, religion 2 first exceeds 1% at year 17.6 and reaches parity with religion 1 at year 50.3. At year 216, mean shares are $y_1 = 0.096$ and $y_2 = 0.890$. Removing mutation leaves religion

2 identically absent. Thus its establishment in this experiment is genuinely schism-mediated, while its subsequent expansion follows from the assigned conversion and retention advantage.

In Scenario B, each minor religion peaks at its externally specified rate-change time. Peak mean shares range from approximately 0.136 to 0.237, and the combined minor share at year 216 is 0.0335. The incumbent ends at 0.274, rather than remaining constant. Removing introductions leaves every minor religion absent; removing the rate changes instead allows the earliest minor religion to persist at a terminal deterministic share of approximately 0.651. The short-lived waves therefore depend on the prescribed schedule. They are neither evidence of autonomous oscillations nor a prediction of movement-specific collapse dates.

In Scenario C, terminal mean shares are 0.461 and 0.456. The symmetric deterministic equilibrium has $y_1 = y_2 = 0.4575$ and all six constrained Jacobian eigenvalues are negative, with largest real part -0.0010 year^{-1}. This supports locally stable coexistence for the specified parameterization. With the much weaker symmetric mutation rate 0.0002, the unequal initialization instead yields deterministic terminal shares of approximately 0.905 and 0.031 at year 1000. Balanced coexistence is therefore a property of the specified coupling and role structure, not a consequence of parameter symmetry alone. The equilibrium also allocates a large fraction to the protected P role; these uncalibrated role proportions should not be read as realistic priesthood prevalence.

4.2.2 Historical descriptive fits and held-out performance

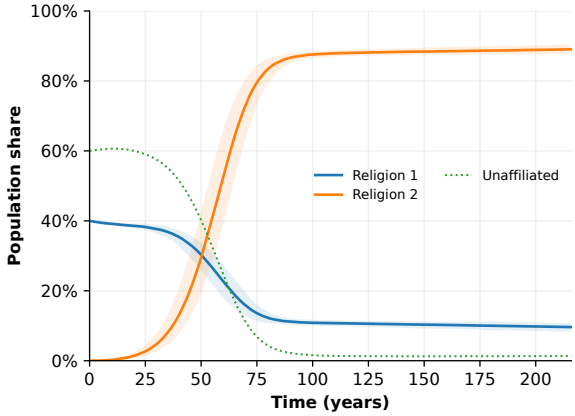
Figure 3 and Table 4 separate fitted agreement from held-out errors. For Sweden, the fixed-break, continuous rate-shift fit improves R^2 from 0.813 to 0.996, and MAE from 5.275 to 0.809 percentage points. The fitted recruitment rates lie at the lower bound in both segments; the effective loss rate increases from approximately 0.00408 to 0.02068 year^{-1}. This is a descriptive summary of declining membership, not evidence that recruitment literally ceased. Training through 2014 gives a held-out MAE of 2.244 percentage points for the rate-shift model, compared with 12.771 for the anchored linear trend.

For Turkey, the logistic description attains $R^2 = 0.918$ and MAE 0.312 peak publishers per 100,000 residents. It summarizes the broad increase but cannot reproduce the observed 2024 decline as an actual downturn. Its held-out MAE is 0.723 publishers per 100,000, compared with 0.325 for the linear trend. Thus the additional saturation parameter does not establish a predictive advantage on this holdout.

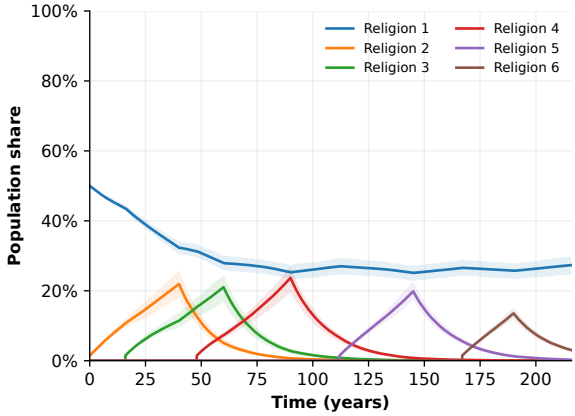
New Zealand's constructed no-religion share increases at every census; it exhibits no rise-and-fall episode. The constant-rate fit has $R^2 = 0.968$ and MAE 1.166 percentage points. Its 2023 holdout error is 1.197 percentage points, versus 0.302 for the linear trend. Fitting the separately published percentage series instead raises in-sample R^2 to approximately 0.993 and changes the estimated rates, illustrating sensitivity to response denominators and data processing. Neither fit validates conversion mechanisms, and near-perfect interpolation of five points is not sought. Table 5 reports every dynamical parameter, including both Sweden periods.

4.2.3 Parameter sensitivity and numerical agreement

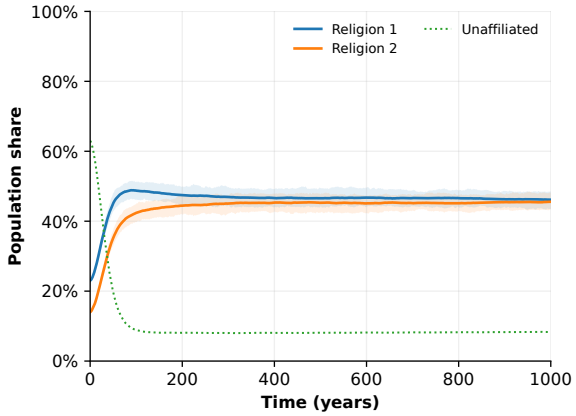
Figure 4 displays continuous finite-horizon changes in religion 2's prevalence as conversion and missionary disaffiliation vary. Greater recruitment and lower loss generally favor expansion in this configuration, but a visible gradient is not itself a phase transition. Doubling the horizon changes



(a) Schism-mediated replacement.



(b) Externally forced transients.

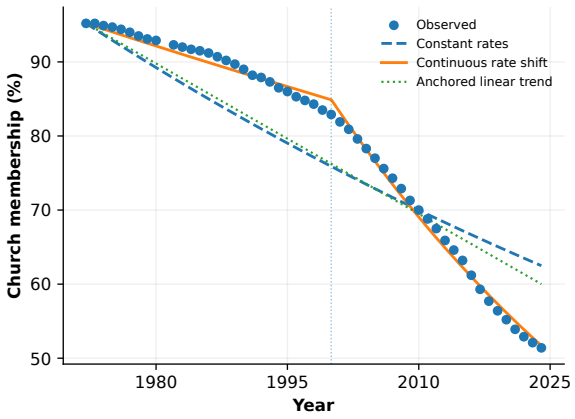


(c) Mutation-supported coexistence.

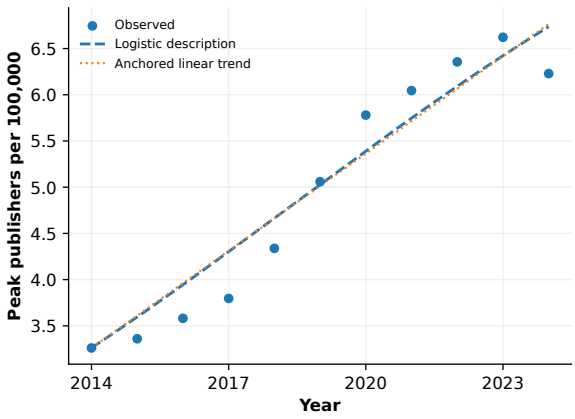
Figure 2: Full-model stochastic simulations (100 replicates per scenario). Lines are replicate-mean shares and shaded bands are pointwise central 95% replicate intervals, not confidence intervals for historical predictions. Scenario B uses the complete forcing schedule in Table 3. Time is in years, with the longer horizon in C chosen for relaxation.

Table 4: Historical fitting and held-out errors. Sweden and NZ MAE values are percentage points; Turkey MAE values are publishers per 100,000 residents. n excludes the fixed initial observation and k counts fitted curve parameters. Holdouts are 2015–2024 (Sweden), 2022–2024 (Turkey), and 2023 (NZ). Lower MAE is better.

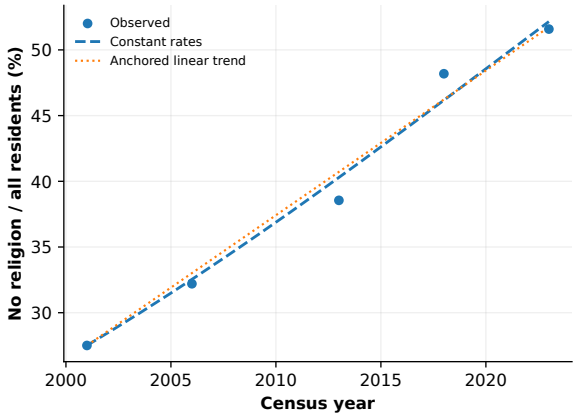
Case	Specification	n	k	In-sample R^2	Fit MAE	Holdout MAE
Sweden	Constant rates	51	2	0.813	5.275	14.327
Sweden	Rate shift	51	4	0.996	0.809	2.244
Sweden	Linear trend	51	1	0.861	4.590	12.771
Turkey	Logistic	10	2	0.918	0.312	0.723
Turkey	Exponential	10	1	0.887	0.336	0.723
Turkey	Linear trend	10	1	0.910	0.330	0.325
NZ	Constant rates	4	2	0.968	1.166	1.197
NZ	Linear trend	4	1	0.961	1.272	0.302



(a) Sweden: membership.



(b) Turkey: peak publishers.



(c) NZ: no-religion reporting.

Figure 3: Observed records and full-sample descriptive fits. The observables differ across cases and are not treated as interchangeable measurements of the full model. NZ uses no-religion counts divided by all census residents. Dashed/dotted comparisons and legends identify the fitted specifications; the Sweden change is continuous at 2000. Held-out forecasts are shown separately in the Appendix.

Table 5: All fitted parameters of the historical dynamical curves. Values below 10^{-8} are shown as zero (boundary solutions). Rate parameters are year^{-1}; K and y_0 are population fractions. The Sweden rate shift is fixed at 2000 and the simulated state remains continuous there.

Case	Specification/period	β	ρ	r	K	y_0
Sweden	Constant	0	0.0080957	–	–	0.952
Sweden	Pre-2000	0	0.0040809	–	–	0.952
Sweden	Post-2000	0	0.020682	–	–	0.952
Turkey	Logistic	–	–	0.15241	9.5816e-05	3.2606e-05
NZ	Constant	0.047966	0	–	–	0.27508

the terminal share by as much as 37.3 percentage points across the grid. It would therefore be misleading to assign universal long-run regimes using a single finite-horizon threshold.

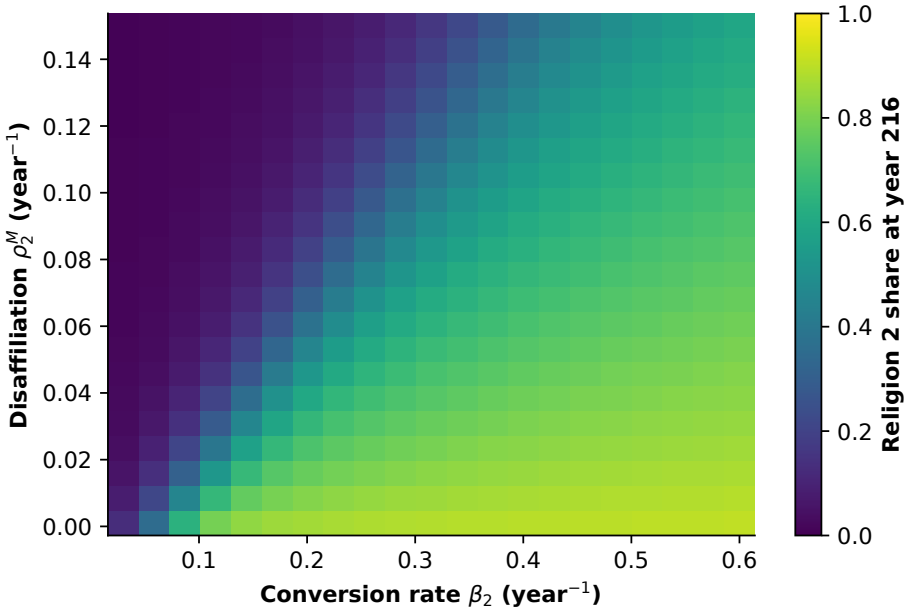


Figure 4: Deterministic finite-horizon sensitivity of Scenario A on a 21×21 grid. Color denotes $y_2(216)$, not a categorical regime or takeover probability. All rates are annual, and other parameters are fixed. The companion data include year-432 values to expose horizon sensitivity.

Across 30 perturbed configurations (ten per scenario), using 20 stochastic replicates per configuration, mean religion-averaged MAE is 0.00163 ± 0.00107 and mean R^2 is 0.99879 ± 0.00181; the uncertainty here is the sample standard deviation across configurations. The respective ranges are [0.00036, 0.00434] and [0.99155, 0.99996]. Figure 5 displays exactly these 30 case-level observations. A and C include mutation; B has none and retains its scheduled introductions and rate changes. These checks concern implementation agreement, not empirical validity.

5 Discussion

In this study, we introduce an epidemiologically inspired framework for modeling religious dynamics as a multi-strain diffusion process with role heterogeneity and schism-like missionary relabeling. Building on classical compartment modeling and cultural-diffusion perspectives, the model formalizes contact-driven conversion, institutional retention, and cross-affiliation competition. We evaluate its internally generated dynamics separately from externally forced experiments and from reduced descriptive fits to historical records. An explicit contextual-rate extension is specified, but no social or political coefficient is estimated in the present study.

Our results show that the proposed structure can represent emergence and replacement, externally induced rise-and-fall trajectories, and mutation-supported coexistence under different settings. These outcomes are consistent with the range of adoption trajectories discussed in diffusion and cultural transmission research [96, 97, 98], without establishing that the selected parameters explain a particular historical mechanism. The ablations sharpen this distinction: Scenario A requires incoming schism to seed an absent challenger; Scenario B requires imposed introductions and subsequent adverse rate changes to produce the displayed waves; and Scenario C's balanced coexistence depends on its coupling strength. In contrast to threshold-based

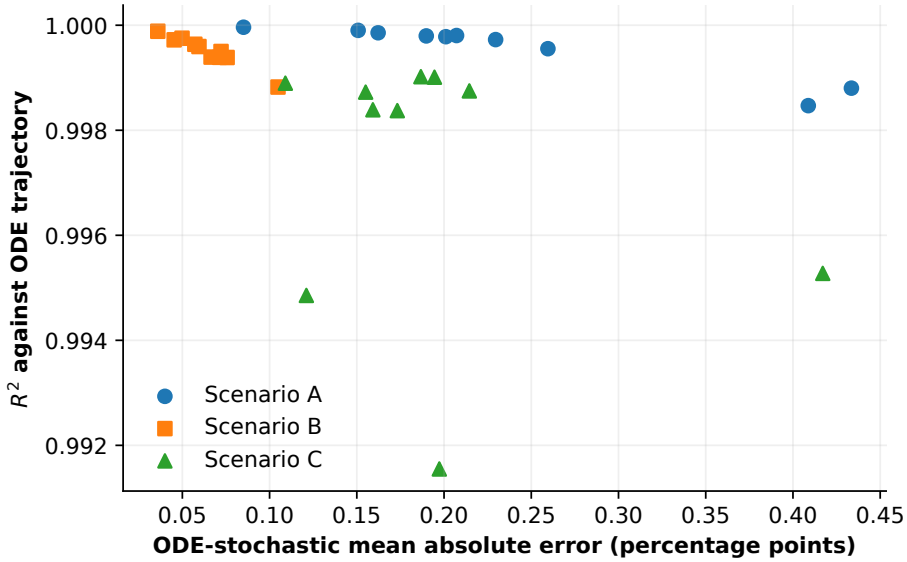


Figure 5: ODE versus stochastic replicate-mean agreement for 30 configurations, with 20 replicates each. Each marker averages MAE and R^2 over that configuration's religions; marker shape identifies the source scenario. MAE on the horizontal axis is expressed in percentage points. Mutation is included in A and C, not B.

complex-contagion models [99, 100], the present homogeneous-mixing conversion hazard is linear in missionary prevalence and contains no social-reinforcement threshold. The finite-horizon sensitivity map therefore demonstrates parameter sensitivity, not complex contagion or a proven phase transition [101].

Interpreted sociologically, the parameters provide a compact language for expressing competition in a shared adherent pool and organizational differences in recruitment and retention [102]. The historical fits show that Sweden's membership decline can be summarized more accurately by a continuous change in effective rates than by a stationary specification. However, the aggregate data do not distinguish interpersonal conversion from cohort replacement, administrative membership changes, or migration. The Turkish publisher and NZ reporting series further demonstrate why descriptive fit should not be equated with forecasting: simpler linear trends achieve lower held-out errors in those two cases. These observations are compatible with, but do not independently test, wider accounts of institutional change and existential security [103, 104, 36].

The framework may support sociological inquiry in three ways. First, competing hypotheses can be mapped to explicit parameter changes and evaluated using observable predictions and ablations, rather than curve resemblance alone. Second, comparative analysis could relate estimated rates to education, state policy, media exposure, or social heterogeneity when harmonized data and identifiability permit [96, 104]. Such an analysis would require measuring the covariates in Eq. (10), specifying an observation model, and testing against appropriate simpler alternatives. Third, the full model provides a conditional simulator for exploring changes in conversion opportunities, retention, or schism propensity. These simulations are mathematical scenarios, not identified historical counterfactuals or policy-effect estimates. This distinction fits the broader use of epidemic-style formalisms for rumors and information diffusion [105, 106, 107, 108, 109].

This study has several limitations. First, homogeneous mixing omits structured networks, spatial heterogeneity, and reinforcement; network or spatial extensions could represent these

processes more directly [99, 110, 111]. Mobility and migration can also interact with religious change, but are not included here [21, 112]. Second, exclusive affiliations omit syncretism and dual practice, and the fixed label set limits the meaning of endogenous innovation. Third, the illustrative settings were selected to explore particular behaviors, not estimated from role-resolved observations. The protected-role accumulation in Scenario C, copying of behavioral roles by the demographic bookkeeping scheme, and the imposed events in B preclude literal demographic interpretation. Fourth, historical fits calibrate reduced curves only; they neither identify the full model nor compare its extra compartments empirically with simpler mechanisms. Low-frequency NZ data, denominator and census-processing differences, organization-reported publisher counts, boundary optima, and small temporal holdouts further restrict inference. Richer observables, cohort and migration data, and an explicit measurement-error model are required before uncertainty in mechanism parameters or cross-context prediction can be assessed reliably [97, 104].

Taken jointly, the proposed model provides a reproducible, hypothesis-generating account of how role-structured conversion, retention, and schism interact in a shared population. Its contribution is a transparent dynamical framework, verified implementation, and a clear separation between endogenous mechanisms, prescribed forcing, and descriptive historical agreement. The results do not establish a universally predictive model of religious change or identify political causes. They instead specify testable mechanisms and the additional measurements needed to evaluate those mechanisms empirically.

Declarations

Funding

This study received no funding.

Data availability

The numerical input series, source definitions, access dates, and provenance notes used in this revision accompany the code. Public sources are cited in the text. All historical observations are aggregate data; no individual-level records are used.

Code availability

The revised reproducibility package accompanies this manuscript. It adapts the publicly available project `https://github.com/bilgesi/religion-dynamics-ode-abs` [90] and includes executable configurations, input CSV files, saved numerical results, figure and table generators, and regression tests. The original project license is retained. All main figure filenames are preserved; the package does not depend on downloading a mutable external dataset at execution time.

Conflicts of interest

The authors declare no conflict of interest.

Author contributions

Bilge Taskin: Software, Formal analysis, Investigation, Data curation, Writing – review and editing, Visualization. Teddy Lazebnik: Conceptualization, Methodology, Validation, Writing – original draft, Writing – review and editing, Visualization, Supervision.

A Computation, diagnostics, and reproducibility

A.1 Exchangeable-agent stochastic computation

The fixed state space contains $1 + 3m$ compartments. Mutation does not make the dimension dynamic and does not prevent ordinary numerical integration. Stochastic simulation is used instead to quantify finite-population variability and discrete seeding, rather than to remedy an ill-defined ODE. This distinction is consistent with comparing equation-based and individual-based computational descriptions [113, 114, 115, 68, 116].

At the start of each step, let $Q_{ij}(x, t)$ be the per-person hazard of transition from state i to a distinct state j, and let an additional destination represent death. Its entries are exactly the conversion, role-change, disaffiliation, mutation, and death rates of Eqs. (6)–(9). For $H_i = \sum_j Q_{ij}$, the number leaving compartment i is sampled as

$$L_i \sim \text{Binomial}\{X_i, 1 - \exp(-H_i \Delta t)\}. \tag{12}$$

Conditional destinations follow a multinomial distribution with weights Q_{ij}/H_i. This enforces at most one competing transition or death per individual per step, rather than independently applying mutually incompatible role changes. Independent births entering i are Poisson$(bX_i\Delta t)$, calculated using start-of-step counts. All changes are applied simultaneously after hazards have been computed. Under homogeneous mixing, this count-based sampler is equivalent to the corresponding exchangeable individual-agent update; it does not introduce a contact network.

The method exactly samples the specified frozen-hazard discrete-time process, but is not an exact continuous-time Markov-chain algorithm. Births and sequential within-step events introduce finite-step approximation relative to the continuous model. Scheduled introductions transfer existing people from S and are applied exactly once at their stated grid times. The ODE is integrated to each actual event endpoint before the transfer or rate change; saved event-time states are right-continuous. Both computations use the same rates, event list, and compartment ordering.

For each replicate we calculate

$$y_r(t) = \frac{B_r(t) + M_r(t) + P_r(t)}{N(t)}, \tag{13}$$

and only then average over replicates. This is not replaced by a ratio of mean counts. The displayed bands are pointwise empirical replicate quantiles. The master random seed is 20261022, scenario seeds are separated by 1000, and per-replicate seeds are derived using NumPy `SeedSequence`; exact seed arrays are saved with the trajectories. No run is selected or omitted on the basis of its outcome.

A.2 ODE–stochastic comparison

The comparison contains exactly 30 configurations: ten perturbed versions of each of A, B, and C, with $T = 216$ years and $\Delta t = 0.1$ throughout. Each nonzero behavioral rate and each q_r is multiplied by an independent draw from Uniform$(0.7, 1.3)$; probabilities are clipped to $[0, 1]$. Nonzero mutation entries are independently perturbed by the same rule, so A and C retain mutation and B retains zero mutation. Initial conditions, demographic rates, and event times are fixed. B's pre- and post-change values of a given religion-specific rate share the same perturbation multiplier. Each configuration uses 20 independent stochastic replicates.

MAE, RMSE, and $R^2 = 1 - \sum_t (y_t^{\mathrm{ABS}} - y_t^{\mathrm{ODE}})^2 / \sum_t (y_t^{\mathrm{ODE}} - \bar{y}^{\mathrm{ODE}})^2$ are calculated per religion on a common grid, using the deterministic series as reference. Case-level values are unweighted means across religions, and reported summary means and standard deviations are across the 30 cases. They are not confidence intervals for a single case. All denominators in these reported R^2 computations are nonzero. The 30 configurations, per-religion metrics, and case-level metrics accompany the figure.

Table 6 compares steps 0.1, 0.05, and 0.025 years on a common annual output grid. Both stochastic sampling and finite-population differences contribute to the discrepancy from the deterministic solution. Accordingly, independent Monte Carlo errors do not decrease monotonically with step size; these checks do not establish equality of a finite-population mean and the nonlinear ODE. Conservation, boundary behavior, absent-religion invariance, event-state continuity, and agreement with an independently written RHS are additionally covered by executable regression tests.

Table 6: Finite-step stochastic checks on a common annual grid over 216 years, using 100 replicates per row. MAE is against the ODE; MCSE is the mean pointwise standard error of the replicate mean. Both are fractions. These independently sampled errors need not decrease monotonically with step size.

Scenario	Δt (years)	MAE	Mean pointwise MCSE
A	0.100	0.001273	0.001076
A	0.050	0.002118	0.000925
A	0.025	0.002269	0.001094
B	0.100	0.000217	0.000423
B	0.050	0.000191	0.000430
B	0.025	0.000264	0.000414
C	0.100	0.000624	0.001337
C	0.050	0.000735	0.001325
C	0.025	0.001438	0.001360

A.3 Coexistence and horizon diagnostics

For Scenario C, solving the frequency equations gives the equilibrium

$$(s, b_1, m_1, p_1, b_2, m_2, p_2) \simeq (0.084976, 0.116574, 0.055359, 0.285579, 0.116574, 0.055359, 0.285579). \tag{14}$$

The constrained Jacobian eliminates $s = 1 - \sum_r (b_r + m_r + p_r)$; its eigenvalues are approximately -0.101992, -0.080837, -0.027345, -0.016912, -0.002723, and -0.001000 year^{-1}. The population-conservation zero direction is therefore not mistakenly included in the stability test.

This numerical local analysis supports the specific coexistence example, not global attraction or robustness of all symmetric settings. The full equilibrium residual, eigenvalues, ablations, and both sensitivity-map horizons are saved without rounding in the code outputs.

B Historical calibration details

B.1 Observation construction and source audit

Table 7: NZ input counts and the constructed all-resident share. Published percentages are retained separately, not substituted for the count ratio. The denominator includes people without a classified affiliation response.

Year	No-religion count	All-resident count	Constructed share (%)	Published (%)
2001	1,028,049	3,737,280	27.508	29.6
2006	1,297,104	4,027,947	32.203	34.6
2013	1,635,345	4,242,048	38.551	41.9
2018	2,264,601	4,699,755	48.186	48.2
2023	2,576,049	4,993,923	51.584	51.6

NZ no-religion counts through 2018 come from the 2019 Stats NZ release; the 2023 count is from the 2024 census release. The 2001 all-resident total is from the government Te Ara historical census table. The 2006 total is the 2013 census count less the officially reported 2006–2013 increase; later totals are reported directly. Random rounding means some releases differ by three persons. We retain the documented release-specific counts rather than silently swapping later rounded versions. The published percentages in Table 7 are a separate sensitivity series because their response denominators and processing differ. They are not alternative estimates obtained merely by rounding the primary ratios. In particular, dividing all five numerators by the 2013 population, or summing overlapping affiliations as a denominator, would not define the series used here.

Swedish observations are the 52 percentages in the cited official membership table, with 1981 omitted rather than interpolated. Turkey observations consistently use the *peak* column; the data package also retains the source population and numerator, making the small share values auditable. Peak activity is an administrative maximum over a service year, not a census of all adherents. The source-year references and the extent of primary-source checking are recorded in `data/provenance.json`.

B.2 Exact solution and identifiable fitting

For a segment satisfying $\dot{y} = ay - cy^2$, the state propagated from its initial value y_0 after duration u is

$$y(u) = \frac{y_0 e^{au}}{1 + (cy_0/a)(e^{au} - 1)}, \quad a \neq 0; \qquad y(u) = \frac{y_0}{1 + cy_0 u}, \quad a = 0. \tag{15}$$

For Eq. (11), $a = \beta - \rho$ and $c = \beta$; for Turkey's logistic curve, $a = r$ and $c = r/K$. Sweden's second segment starts at the first segment's predicted value at 2000. No observed breakpoint value enters as a second initial condition. Linear and exponential alternatives are $y_0 + a(t - t_0)$ and $y_0 e^{g(t-t_0)}$, respectively. Their fitted values remain within the share range over the evaluated periods; they are descriptive baselines, not bounded population models for unrestricted extrapolation.

Least squares uses residuals after the fixed initial observation and scales them by the training-series range only to improve numerical conditioning. Bounds are $0 \le \beta, \rho \le 1$ year^{-1} in each relevant segment, $10^{-8} \le r \le 1$ year^{-1}, and $1.000001 \max(y_{\text{train}}) \le K \le 1$. Logistic optimization uses $\log K$ internally but reports K. The exponential rate is bounded to $[-1, 1]$ year^{-1}, and the anchored linear slope has its closed-form least-squares solution. Nonlinear optimization uses SciPy `least_squares`, tolerances 10^{-12}, and up to 2500 function evaluations per start. Starting values, bounds, successful-fit flags, and residual-Jacobian singular values are saved. Multiple starts are a robustness check, not a mathematical guarantee of a global optimum.

The Swedish recruitment estimates and the primary NZ loss estimate are at their nonnegative lower bounds. They must not be interpreted as measured zero conversion or zero return to religion. Aggregate shares confound omitted mechanisms, and small residual errors do not identify the full role structure. For the descriptive AICc diagnostic we use

$$\text{AICc} = n \log(\text{SSE}/n) + 2k^* + \frac{2k^*(k^*+1)}{n - k^* - 1}, \qquad k^* = k + 1, \tag{16}$$

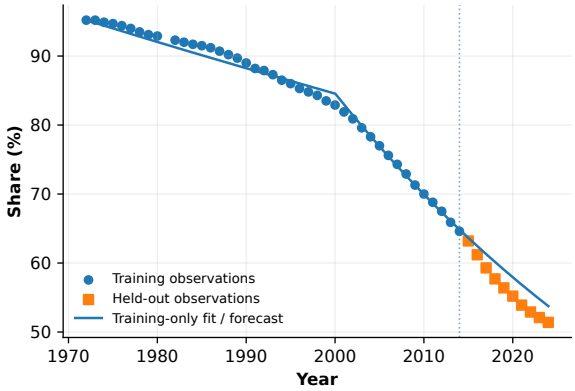
where k is the number of curve parameters, the extra parameter is an independent Gaussian error variance, and n excludes initialization. Constants common to models fitted to the same observations are omitted. This working error model is not a substantive model of census uncertainty or temporal dependence. For the two-parameter NZ curve, $n = 4$ makes the correction undefined; no finite AICc is reported. Table 8 also supplies the fitted trend coefficients.

Table 8: Descriptive complexity diagnostics and anchored trend coefficients. AICc assumes independent Gaussian errors, counts the error variance as a parameter, and is not reported when the small-sample correction is undefined. Comparisons across different datasets are not meaningful. a is the anchored linear slope (fraction/year), and g is the exponential rate (year^{-1}).

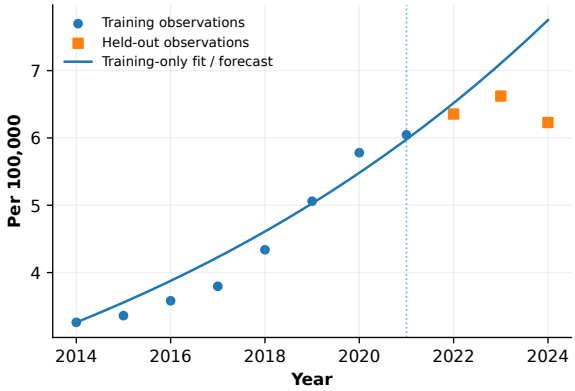
Case	Specification	AICc	Trend coefficient
Sweden	Constant rates	-280.992	–
Sweden	Rate shift	-468.808	–
Sweden	Linear trend	-298.236	$a = -0.00676724$
Turkey	Logistic	-241.83	–
Turkey	Exponential	-242.871	$g = 0.076951$
Turkey	Linear trend	-245.107	$a = 3.50551e-06$
NZ	Constant rates	–	–
NZ	Linear trend	-17.498	$a = 0.0110103$

B.3 Reproduction

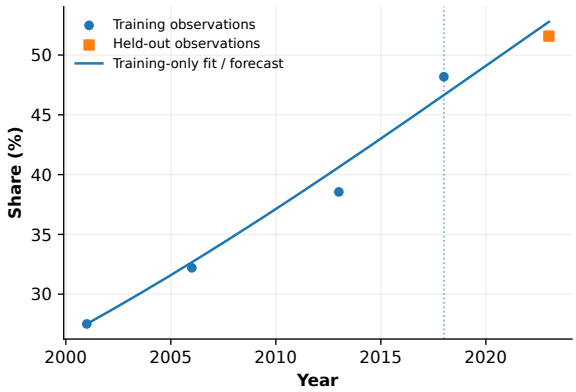
The package includes a version-pinned Python environment, configuration JSON files, a source and data hash manifest, the 17-test regression suite, and all numerical CSV/NPZ/JSON outputs. Run `python run_all.py`, followed by `python make_paper_assets.py`, from the code directory; the latter regenerates numerical LaTeX macros and tables. The schematic has an included vector TikZ source. Compile `main.tex` with PDFLaTeX and BibTeX. No historical point is synthesized and no result is copied from the earlier inconsistent tables. The inspected upstream source files and the precise changes from the public implementation are documented in the accompanying audit notes.



(a) Sweden; train through 2014.



(b) Turkey; train through 2021.



(c) NZ; train through 2018.

Figure 6: Genuine held-out evaluations of the selected reduced dynamical specifications. The curves use training-only parameter estimates on both sides of the vertical cutoff. Square points were not used for fitting, breakpoint selection, or state reinitialization. Full-data fits in Figure 3 are not substituted for these forecasts.

References

[1] Mark Juergensmeyer. *Religion in global civil society*. Oxford University Press, 2005.

[2] Meredith B McGuire. *Religion: The social context*. Waveland Press, 2008.

[3] Robert N Bellah. *Beyond belief: Essays on religion in a post-traditionalist world*. Univ of California Press, 1991.

[4] Martin Lang. The evolutionary paths to collective rituals: An interdisciplinary perspective on the origins and functions of the basic social act. *Archive for the Psychology of Religion*, 41(3):224–252, 2019.

[5] Martin D Stringer. Building on belief: Defining animism in tylor and contemporary society. In *The handbook of contemporary animism*, pages 63–72. Routledge, 2014.

[6] Lance Grande. *The evolution of religions: A history of related traditions*. Columbia University Press, 2024.

[7] Tammi J Schneider. *An introduction to ancient Mesopotamian religion*. Wm. B. Eerdmans Publishing, 2011.

[8] Emily Teeter. *Religion and ritual in ancient Egypt*. Cambridge University Press, 2011.

[9] Upinder Singh. *The Idea of Ancient India: Essays on Religion, Politics and Archaeology*. Penguin Random House India Private Limited, 2023.

[10] Carlton H Tucker. From the axial age to the new age: Religion as a dynamic of world history. *The History Teacher*, 27(4):449–464, 1994.

[11] Jan Assmann. World religions and the theory of the axial age. *Dynamics in the History of Religions*, 1, 2012.

[12] Abbas Mirakhor and Hossein Askari. Conception of justice: Axial age—india, china and greece. In *Conceptions of Justice from Earliest History to Islam*, pages 115–164. Springer, 2019.

[13] Dorothea Weltecke. The medieval period. In *The Oxford Handbook of Atheism*, pages 164–178. Oxford University Press Oxford, 2013.

[14] Clifford Hugh Lawrence and Janet Burton. *Medieval monasticism: forms of religious life in Western Europe in the Middle Ages*. Routledge, 2023.

[15] Edward Grant. *Science and Religion, 400 BC to AD 1550: from Aristotle to Copernicus*. JHU Press, 2006.

[16] Tuomas Martikainen. The global political economy, welfare state reforms, and the governance of religion. In *Post-Secular Society*, pages 71–93. Routledge, 2017.

[17] François Gauthier. *Religion in the neoliberal age: Political economy and modes of governance*. Routledge, 2016.

[18] Peter L Berger. The desecularization of the world: A global overview. In *The new sociology of knowledge*, pages 61–76. Routledge, 2017.

[19] Gabriel A Almond, R Scott Appleby, and Emmanuel Sivan. *Strong religion: The rise of fundamentalisms around the world*. University of Chicago Press, 2011.

[20] Nikki R Keddie. The new religious politics: where, when, and why do "fundamentalisms" appear? *Comparative studies in society and history*, 40(4):696–723, 1998.

[21] Manuel A Vásquez. Religion, globalization, and migration. In *Religions in the Modern World*, pages 431–452. Routledge, 2016.

[22] Ruth Tsuria and Aya Yadlin-Segal. Digital religion and global media: Flows, communities, and radicalizations. In *Handbook of global media ethics*, pages 157–175. Springer, 2021.

[23] Michael Wilkinson. The emergence, development, and pluralisation of global pentecostalism. In *Handbook of Global Contemporary Christianity*, pages 93–112. Brill, 2015.

[24] Enzo Pace. The catholic charismatic movement in global pentecostalism. *Religions*, 11(7):351, 2020.

[25] Allan Anderson, Michael Bergunder, Andre F Droogers, and Cornelis Van der Laan. *Studying global Pentecostalism: Theories and methods*, volume 10. Univ of California Press, 2010.

[26] Clive D Field. *Counting religion in Britain, 1970-2020: secularization in statistical context*. Oxford University Press, 2021.

[27] Amr Sabet. Religion, politics and social change: A theoretical framework. *Religion, State and Society: The Keston Journal*, 24(2-3):241–268, 1996.

[28] Mehmet Karaçuka. Religion and economic development in history: Institutions and the role of religious networks. *Journal of Economic Issues*, 52(1):57–79, 2018.

[29] Robert L Montgomery. The spread of religions and macrosocial relations. *Sociological Analysis*, 52(1):37–53, 1991.

[30] Robert L Montgomery. Special persons and the spread of religions. *Pastoral psychology*, 65(3):369–393, 2016.

[31] Safina Safitri. The role of social media in spreading religious moderation values. *Young Journal of Social Sciences and Humanities*, 1(1):99–107, 2025.

[32] Augustine WC Msiska. The spread of islam in malawi and its impact on yao rites of passage, 1870-1960. *The Society of Malawi Journal*, 48(1):49–86, 1995.

[33] John Hayward. Mathematical modeling of church growth. *The Journal of mathematical sociology*, 23(4):255–292, 1999.

[34] M. Afzal Upal. Simulating the emergence of new religious movements. *Journal of Artificial Societies and Social Simulation*, 8(1):6, 2005.

[35] Michael Doebeli and Iaroslav Ispolatov. A model for the evolutionary diversification of religions. *Journal of Theoretical Biology*, 267(4):676–684, 2010.

[36] Ross J. Gore, Carlos Lemos, F. LeRon Shults, and Wesley J. Wildman. Forecasting changes in religiosity and existential security with an agent-based model. *Journal of Artificial Societies and Social Simulation*, 21(1):4, 2018.

[37] Matteo Castiello, Dajana Conte, and Samira Iscaro. Using epidemiological models to predict the spread of information on twitter. *Algorithms*, 16(8), 2023.

[38] W. O. Kermack and A. G. McKendrick. A contribution to the mathematical theory of epidemics. *Proceedings of the Royal Society*, 115:700–721, 1927.

[39] Michael A Cretacci. Religion and social control: An application of a modified social bond on violence. *Criminal Justice Review*, 28(2):254–277, 2003.

[40] Edward Alsworth Ross. Social control. v. religion. *American journal of sociology*, 2(3):433–445, 1896.

[41] Anneli Kaasa. Religion and social capital: Evidence from european countries. *International Review of Sociology*, 23(3):578–596, 2013.

[42] Mircea Eliade. *A history of religious ideas volume 1: From the Stone age to the Eleusinian mysteries*, volume 1. University of Chicago Press, 2020.

[43] Richard Sosis and Bradley J Ruffle. Ideology, religion, and the evolution of cooperation: Field experiments on israeli kibbutzim. In *Socioeconomic aspects of human behavioral ecology*, pages 89–117. Emerald Group Publishing Limited, 2004.

[44] Joseph Bulbulia. Spreading order: religion, cooperative niche construction, and risky coordination problems. *Biology & Philosophy*, 27(1):1–27, 2012.

[45] Andrew Phillips. *War, religion and empire: The transformation of international orders*, volume 117. Cambridge University Press, 2010.

[46] Richard Sosis. Does religion promote trust?: The role of signaling, reputation, and punishment. *Interdisciplinary journal of research on religion*, 1, 2005.

[47] Gary D Bouma. Religious resurgence, conflict and the transformation of boundaries. In *Religion, Globalization, and Culture*, pages 187–202. Brill, 2007.

[48] Andreas Hasenclever and Volker Rittberger. Does religion make a difference? theoretical approaches to the impact of faith on political conflict. *Millennium*, 29(3):641–674, 2000.

[49] Wade Clark Roof. Religion and narrative. *Review of religious research*, 34(4):297–310, 1993.

[50] Yosef Garfinkel. *Dancing at the Dawn of Agriculture*. University of Texas Press, 2010.

[51] Dana L Haggard and Stephen Haggard. Governance, law, religion and culture. *International Journal of Organization Theory & Behavior*, 13(4):569–596, 2010.

[52] Richard Foltz. *Religions of the Silk Road*. St. Martin's Press, 1999.

[53] Johan Elverskog. *Buddhism and Islam on the silk road*. University of Pennsylvania press, 2011.

[54] Roger Finke. The consequences of religious competition: Supply-side explanations for religious change. In *Rational choice theory and religion*, pages 46–65. Routledge, 2016.

[55] Jonathan Fox. A world survey of secular-religious competition: state religious policy from 1990 to 2014. In *The Correlates of Religion and State*, pages 9–28. Routledge, 2020.

[56] Barbara Diefendorf. Were the wars of religion about religion? *political theology*, 15(6):552–563, 2014.

[57] Norman Housley. *Religious warfare in Europe 1400-1536*. OUP Oxford, 2008.

[58] Eileen Barker. Revision and diversification in new religious movements: an introduction. In *Revisionism and diversification in new religious movements*, pages 1–14. Routledge, 2016.

[59] James R Lewis and Sarah M Lewis. *Sacred schisms: how religions divide*. Cambridge University Press, 2009.

[60] J Gordon Melton and James R Lewis. Birth of a religion. In *Scientology*, pages 17–33. Oxford University Press Oxford, 2009.

[61] Lu Tang, Yiwang Zhou, Lili Wang, Soumik Purkayastha, Leyao Zhang, Jie He, Fei Wang, and Peter X-K Song. A review of multi-compartment infectious disease models. *International Statistical Review*, 88(2):462–513, 2020.

[62] Sheryl L Chang, Mahendra Piraveenan, Philippa Pattison, and Mikhail Prokopenko. Game theoretic modelling of infectious disease dynamics and intervention methods: a review. *Journal of biological dynamics*, 14(1):57–89, 2020.

[63] Philip E Paré, Carolyn L Beck, and Tamer Başar. Modeling, estimation, and analysis of epidemics over networks: An overview. *Annual Reviews in Control*, 50:345–360, 2020.

[64] Teddy Lazebnik. Computational applications of extended SIR models: A review focused on airborne pandemics. *Ecological Modelling*, 483:110422, 2023.

[65] Teddy Lazebnik. Evaluating supply chain resilience during pandemic using agent-based simulation. *Physica A: Statistical Mechanics and its Applications*, page 130780, 2025.

[66] Cliff C Kerr, Robyn M Stuart, Dina Mistry, Romesh G Abeysuriya, Katherine Rosenfeld, Gregory R Hart, Rafael C Núñez, Jamie A Cohen, Prashanth Selvaraj, Brittany Hagedorn, et al. Covasim: an agent-based model of covid-19 dynamics and interventions. *PLOS Computational Biology*, 17(7):e1009149, 2021.

[67] Petrônio CL Silva, Paulo VC Batista, Hélder S Lima, Marcos A Alves, Frederico G Guimarães, and Rodrigo CP Silva. Covid-abs: An agent-based model of covid-19 epidemic to simulate health and economic effects of social distancing interventions. *Chaos, Solitons & Fractals*, 139:110088, 2020.

[68] Elizabeth Hunter, Brian Mac Namee, and John Kelleher. A hybrid agent-based and equation based model for the spread of infectious diseases. *Journal of Artificial Societies and Social Simulation*, 23(4):14, 2020.

[69] Wenxuan Li, Xu Chen, Suli Liu, Chiyu Zhang, and Guyue Liu. Using a multi-strain infectious disease model with physical information neural networks to study the time dependence of SARS-CoV-2 variants of concern. *PLOS Computational Biology*, 21(2):e1012778, 2025.

[70] Peijue Zhang, Kairui Feng, Yuqing Gong, Jieon Lee, Sara Lomonaco, and Liang Zhao. Usage of compartmental models in predicting covid-19 outbreaks. *The AAPS Journal*, 24(5):98, 2022.

[71] Fatemeh Mirsaeedi, Mohammad Sheikhalishahi, Mehrdad Mohammadi, Amir Pirayesh, and Dmitry Ivanov. Compartmental models in epidemiology: bridging the gap with operations research for enhanced epidemic control. *Annals of Operations Research*, pages 1–58, 2025.

[72] Herbert W. Hethcote. The mathematics of infectious diseases. *SIAM Review*, 42(4):599–653, 2000.

[73] Adi Shuchami and Teddy Lazebnik. Spatio-temporal sir model of pandemic spread during warfare with optimal dual-use health care system administration using deep reinforcement learning. *Disaster Medicine and Public Health Preparedness*, 19:e197, 2025.

[74] Labib Shami and Teddy Lazebnik. Economic aspects of the detection of new strains in a multi-strain epidemiological–mathematical model. *Chaos, Solitons & Fractals*, 165:112823, 2022.

[75] Teddy Lazebnik and Svetlana Bunimovich-Mendrazitsky. Generic approach for mathematical model of multi-strain pandemics. *PloS one*, 17(4):e0260683, 2022.

[76] Viggo Andreasen, Jane Lin, and Simon A. Levin. The dynamics of cocirculating influenza strains conferring partial cross-immunity. *Journal of Mathematical Biology*, 35:825–842, 1997.

[77] Julia R. Gog and Bryan T. Grenfell. Dynamics and selection of many-strain pathogens. *Proceedings of the National Academy of Sciences*, 99(26):17209–17214, 2002.

[78] Pavlo Minayev and Neil M. Ferguson. Improving the realism of deterministic multi-strain models: implications for modelling influenza a. *Journal of the Royal Society Interface*, 6(35):509–518, 2009.

[79] M. Gabriela M. Gomes, Lisa J. White, and Graham F. Medley. The reinfection threshold. *Journal of Theoretical Biology*, 236(1):111–113, 2005.

[80] Amélia Nunes, M. Margarida Telo da Gama, and M. Gabriela M. Gomes. Localized contacts between hosts reduce pathogen diversity. *Journal of Theoretical Biology*, 241(2):477–487, 2006.

[81] Martin Fudolig and Robert Howard. The local stability of a modified multi-strain sir model for emerging viral strains. *PLOS ONE*, 15(12):e0243408, 2020.

[82] Laith J. Abu-Raddad and Neil M. Ferguson. The impact of cross-immunity, mutation and stochastic extinction on pathogen diversity. *Proceedings of the Royal Society of London. Series B: Biological Sciences*, 271(1556):2431–2438, 2004.

[83] Isabel Gordo, M. Gabriela M. Gomes, D. G. Reis, and P. R. A. Campos. Genetic diversity in the sir model of pathogen evolution. *PLOS ONE*, 4(3):e4876, 2009.

[84] Marie T. Meehan, Adam M. Rojek, Adegboye I. Adekunle, Olumuyiwa A. Adegboye, and Emma S. McBryde. Coupled, multi-strain epidemic models of mutating pathogens. *Mathematical Biosciences*, 296:1–17, 2018.

[85] Teddy Lazebnik, Svetlana Bunimovich-Mendrazitsky, Shai Ashkenazi, Eugene Levner, and Arriel Benis. Early detection and control of the next epidemic wave using health communications: development of an artificial intelligence-based tool and its validation on covid-19 data from the us. *International Journal of Environmental Research and Public Health*, 19(23):16023, 2022.

[86] Svenska kyrkan. Svenska kyrkans medlemsutveckling år 1972–2024. PDF report, Svenska kyrkan i siffror (statistik), 2024. Accessed 2026-01-09.

[87] Watch Tower Bible and Tract Society of Pennsylvania. 2024 service year report of jehovah's witnesses worldwide. Web report, 2024.

[88] Watch Tower Bible and Tract Society of Pennsylvania. 2024 country and territory reports. Web table (part of the 2024 Service Year Report), 2024.

[89] Watch Tower Bible and Tract Society of Pennsylvania. Service-year country and territory reports, 2014–2024: Turkey. Annual yearbooks and service-year reports. Peak publisher and population columns; year-specific URLs and verification coverage in the accompanying data provenance file. Accessed 22 October 2026.

[90] Bilge Taskin and Teddy Lazebnik. Religion dynamics ODE–ABS: public computational implementation. GitHub repository. Public source files inspected 22 October 2026; revised reproducibility package accompanies this manuscript.

[91] Stats NZ. Losing our religion, 2019. 3 October 2019. Historical no-religion counts and published percentages; accessed 22 October 2026.

[92] Stats NZ. 2023 census population, dwelling, and housing highlights, 2024. 2023 usually resident population and no-religion count; accessed 22 October 2026.

[93] Manatu Taonga – Ministry for Culture and Heritage. Census population, 1858–2006: underlying data. Te Ara – The Encyclopedia of New Zealand. 2001 census count; accessed 22 October 2026.

[94] Stats NZ. 2013 census usually resident population counts, 2013. 2013 total and increase since 2006; accessed 22 October 2026.

[95] Stats NZ. 2018 census population and dwelling counts, 2019. Usually resident population; accessed 22 October 2026.

[96] Everett M. Rogers. *Diffusion of Innovations*. Free Press, New York, 5 edition, 2003.

[97] Luigi Luca Cavalli-Sforza and Marcus W. Feldman. *Cultural Transmission and Evolution: A Quantitative Approach*. Princeton University Press, Princeton, NJ, 1981.

[98] Robert Boyd and Peter J. Richerson. *Culture and the Evolutionary Process*. University of Chicago Press, Chicago, IL, 1985.

[99] Damon Centola and Michael Macy. Complex contagions and the weakness of long ties. *American Journal of Sociology*, 113(3):702–734, 2007.

[100] Daniel Reisinger, Fabian Tschofenig, Raven Adam, Marie Lisa Kogler, Manfred Füllsack, Fabian Veider, and Georg Jäger. Patterns of stability in complex contagions. *Journal of Computational Social Science*, 7:1895–1911, 2024.

[101] Manfred Füllsack, Simon Plakolb, and Georg Jäger. Predicting regime shifts in social systems modelled with agent-based methods. *Journal of Computational Social Science*, 4:163–185, 2021.

[102] Laurence R. Iannaccone. The consequences of religious market structure: Adam smith and the economics of religion. *Rationality and Society*, 3(2):156–177, 1991.

[103] Peter L. Berger. *The Sacred Canopy: Elements of a Sociological Theory of Religion*. Doubleday, Garden City, NY, 1967.

[104] Pippa Norris and Ronald Inglehart. *Sacred and Secular: Religion and Politics Worldwide*. Cambridge University Press, Cambridge, UK, 2012.

[105] D. J. Daley and D. G. Kendall. Stochastic rumours. *IMA Journal of Applied Mathematics*, 1(1):42–55, 1965.

[106] KM Ariful Kabir, Kazuki Kuga, and Jun Tanimoto. Analysis of sir epidemic model with information spreading of awareness. *Chaos, Solitons & Fractals*, 119:118–125, 2019.

[107] Laijun Zhao, Hongxin Cui, Xiaoyan Qiu, Xiaoli Wang, and Jiajia Wang. Sir rumor spreading model in the new media age. *Physica A: Statistical Mechanics and its Applications*, 392(4):995–1003, 2013.

[108] Xiaobin Rui, Fanrong Meng, Zhixiao Wang, Guan Yuan, and Changjiang Du. Spir: The potential spreaders involved sir model for information diffusion in social networks. *Physica A: Statistical Mechanics and Its Applications*, 506:254–269, 2018.

[109] Miyuki Matsumoto, Ryuji Hashimoto, Masahiro Suzuki, Yuri Murayama, and Kiyoshi Izumi. Impact of information disparity between individual investors on profits of meme stocks using an artificial market simulation approach. *Journal of Computational Social Science*, 8(25), 2025.

[110] QQ Zheng, JW Shen, V Pandey, YM Zhao, and LN Guan. Spatiotemporal dynamics of periodic waves in sir model with driving factors. *New Journal of Physics*, 25(6):063028, 2023.

[111] Mateen Mahmood, André Victor Ribeiro Amaral, Jorge Mateu, and Paula Moraga. Modeling infectious disease dynamics: Integrating contact tracing-based stochastic compartment and spatio-temporal risk models. *Spatial Statistics*, 51:100691, 2022.

[112] Ka U Ng. Are muslim immigrants assimilating? cultural assimilation trajectories in immigrants' attitudes toward gender roles in europe. *Journal of Ethnic and Migration Studies*, 48(15):3641–3667, 2022.

[113] Charles M Macal. To agent-based simulation from system dynamics. In *Proceedings of the 2010 winter simulation conference*, pages 371–382. IEEE, 2010.

[114] Pedro HT Schimit. Multi-strain epidemic models of mutating viruses with airborne transmission based on cellular automata and ordinary differential equations. *Axioms*, 12(7):632, 2023.

[115] Anna Paula Galvão Scheidegger and Amarnath Banerjee. A comparison between a traditional seir-sei epidemiological model and a multi-strain, multi-vector disease spread model. In *IISE Annual Conference. Proceedings*, pages 2181–2186. Institute of Industrial and Systems Engineers (IISE), 2018.

[116] Teddy Lazebnik and Avner Friedman. Comparing partial differential equations and agent-based simulations in spatio-temporal modeling of cancer growth and shape. *Journal of Computational and Applied Mathematics*, page 117183, 2025.